

Annual Review for



LFENERGY

SEAPATH

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SEAPATH

High availability real-time hypervisor for hosting digital substation virtualized protection and control applications (vPAC)

Description

SEAPATH, Software Enabled Automation Platform and Artifacts (Therein), aims at developing a reference design and industrial-grade open source real-time platform that can run virtualized automation and protection applications (for the power grid industry initially, and potentially beyond). This platform is intended to host multi-provider applications.

SEAPATH meets the high requirements of substation reliability, performance, and availability.

Key features include:

- Hosting of virtualization systems: a heterogeneous variety of virtual machines can be installed and managed on the platform.
- High availability and clustering: machines on the cluster are externally monitored in order to guarantee high availability in case of hardware or software failures.
- Distributed storage: data and disk images from the virtual machines are replicated and synchronized in order to guarantee their integrity and availability on the cluster.
- Intelligent virtual network: the virtualisation platform is capable of configuring and managing the network traffic at the data layer level.
- Administration: the system can be easily configured and managed from a remote machine connected to the network, as well as by an administrator on site.
- Automatic update: the virtualisation platform can be automatically updated from a remote server.

Links



Mailing list

<https://lists.lfenergy.org/g/SEAPATH>



Slack

<https://lfenergy.slack.com/archives/C01EH8ZLJTC>



Linkedin

<https://www.linkedin.com/showcase/seapath-project/>



Github

<https://github.com/seapath>



Technical Steering Committee (TSC)

<http://calendar.lfenergy.org>



Documentation

<https://lf-energy.atlassian.net/wiki/spaces/SEAP/overview>

Organizations contributing and/or evaluating in lab/pre-production and/or using in production

Utilities



vIED Vendors



HW Vendors



Software & Integrators



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→ More information on our Wiki : [Who is using SEAPATH ?](#)

Key Achievements in the past year [1/]



V2.0 IS LIVE!



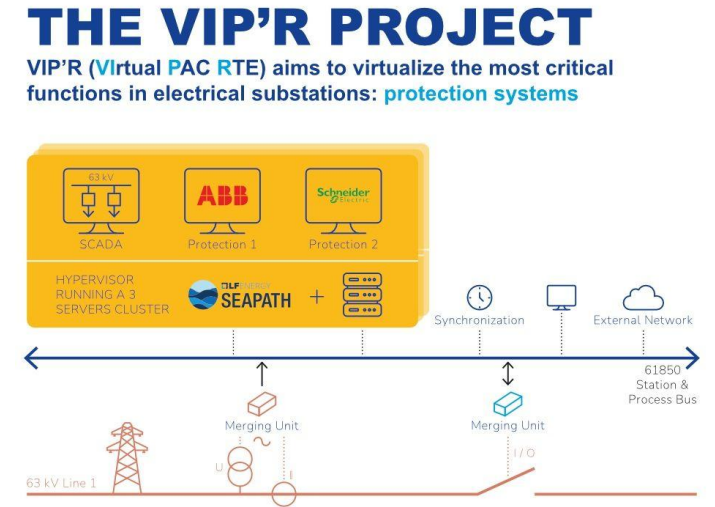
- Release of SEAPATH 2.0
- Transfert to LF Europe voted by TSC
- SEAPATH Documentation refactoring
 - still on [confluence](#)
- SEAPATH Roadmap Workshop at RTE
 - 45 people from 19 companies, 8 countries



➔ **A Strong & Diverse Ecosystem: Utilities, vIED & PAC Vendors, HW/SW vendors, integrators...**

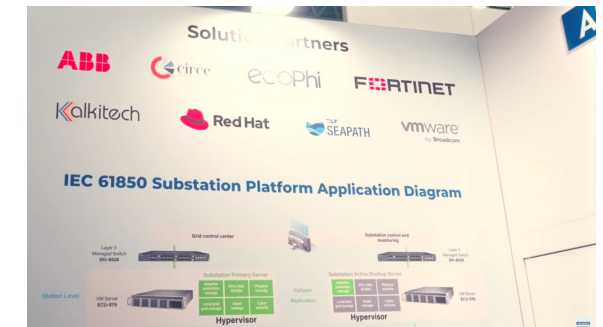
Key Achievements in the past year [2/]

- Industrial collaboration
 - VIP'R project with RTE, Schneider and ABB
 - Support of SEAPATH on Qualcomm IQ9
- Support of enterprise-grade Linux
 - **Suse support** with SLES 16
 - Redhat preliminary support with Centos
- Deployment and Field Use (SEAPATH in production)
 - RTE (**6 Substations** in operation, 4 more by EOY)
 - Ecophi (**22 SEAPATH instances** running in Substations)
 - Labs by Alliander, Elia, Avangrid...



Key Achievements in the past year [3/]

- LF Energy summit 2025 - AAachen - **7 talks** !
 - SEAPATH: What's New - Recent Improvements and Future Developments | Eloi Bail, Savoir-faire Linux
 - What Do Virtualization and SEAPATH Really Change for RTE ? | Bastien Desbos, RTE
 - A Case Study on Testing LFE Seapath at National Grid Electricity Transmission | Thomas Charton, National Grid Electricity Transmission
 - Streamlining SEAPATH Deployment and Management in Electrical Substations | Matias Vara, Red Hat & Erwann Roussy, Savoir-faire Linux
 - Digital Sovereignty and Grid Resilience: The Strategic Role of LF Energy and Open Source | Christophe Villemer, Savoir-faire Linux
 - SBOM & Vulnerabilities in SEAPATH | Mathieu Dupré, Savoir-faire Linux
 - End-to-End Substation Digitalisation with QMU 800 & CMPC 800 | Jakob Lindqvist, Ecophi
- SEAPATH **All over the World**
 - ENLIT Europe (Bilbao, Spain)
 - Open Source Summit Japan (Tokyo, Japan)
 - LF Europe Member Summit (Ghent, Belgium),
 - Distributech (San Diego, CA)
 - Smart Grid Technical Forum SGT (Paris, France)
 - CIGRE International (Montreal, Canada)
 - Open Source Summit North America (Denver, Co)



SEAPATH from Early Adoption to Graduated [1/]

- SEAPATH in Early adoption **since March 2023** ✓
- Achievement of the OpenSSF Best Practices badge at the [‘Gold’ level](#): **87% done**
- Have fulfilled or are on track to complete the growth plan defined in the Early Adoption stage proposal. ✓
- Have a healthy number of contributions or committers from at least three organizations, with any single organization not composing more than 50% of the contributions or committers. Committers must be identified within the project in a COMMITTERS file. ✓
- Have a public list of project adopters for at least the primary repo (e.g., ADOPTERS.md or logos on the project website). ✓

SEAPATH from Early Adoption to Graduated [2/]

- Achievement of the OpenSSF Best Practices badge at the [‘Gold’ level](#).
 - **87% done**
 - Discussion on
 - Working build system: The project MUST have a reproducible build
 - Automated test suite: work to be done on
 - Ansible (Molecule)
 - python3-setup-ovs
 - vm-manager
 - cockpit plugins

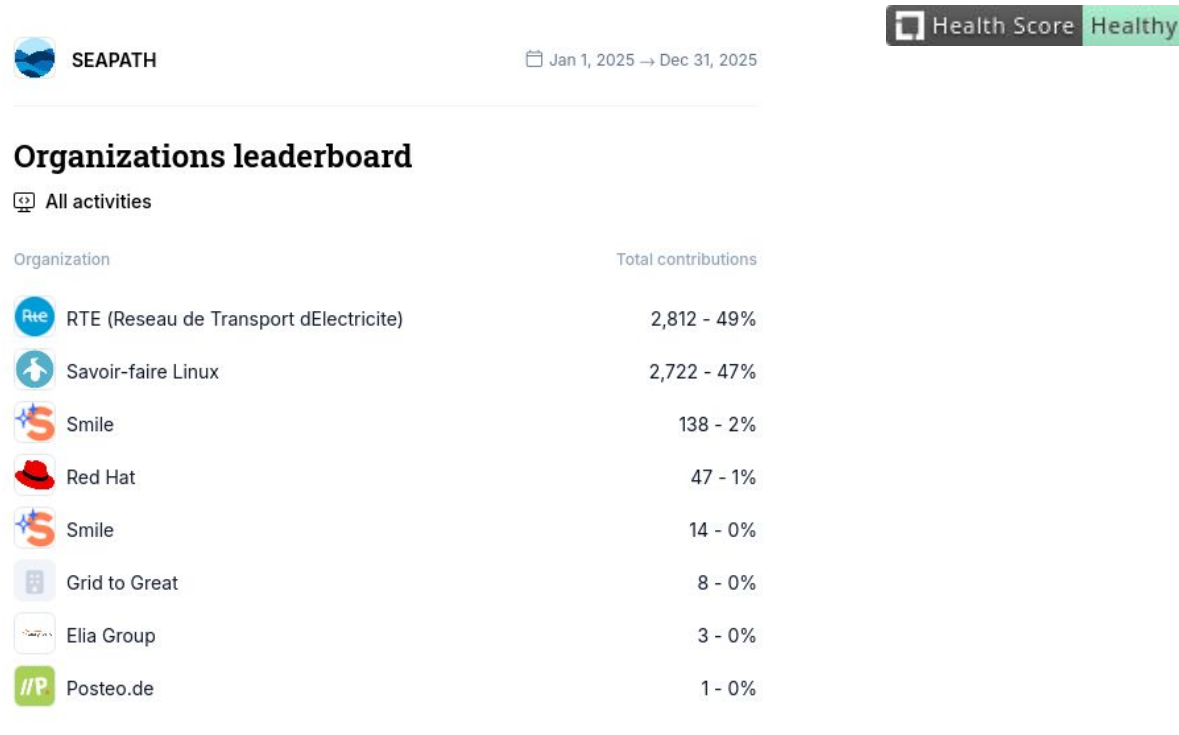
SEAPATH from Early Adoption to Graduated [3/]

- Have fulfilled or are on track to complete the growth plan defined in the Early Adoption stage proposal. ✓
 - Communication around SEAPATH v.1.0 ✓
 - Lot of communication & talks on SEAPATH 1.X & 2.X
 - Individual & group discussion with SEAPATH TSC voting members ✓
 - More companies joining TSC and Roadmap workshop
 - Third party integration ✓
 - Integration with HW manufacturers, SW providers and services companies
 - Attract more utilities ✓
 - Alliander, Elia, Hydro-Québec, National Grid

SEAPATH from Early Adoption to Graduated [4/]

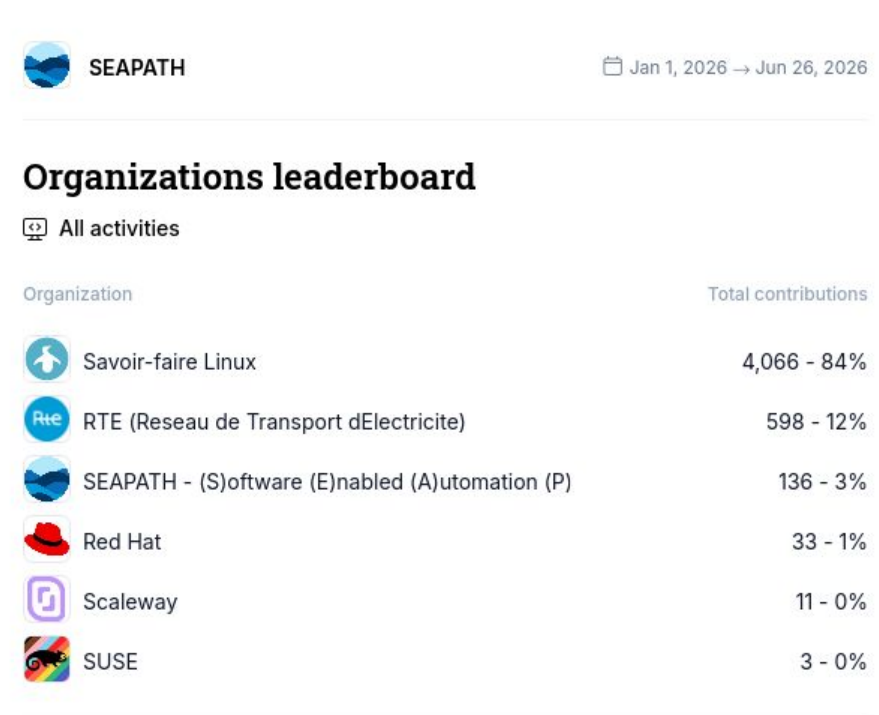
- Have a healthy number of contributions or committers from at least three organizations, with any single organization not composing more than 50% of the contributions or committers. Committers must be identified within the project in a COMMITTERS file. 
 - <https://github.com/seapath/.github/blob/main/COMMITTERS.md>

SEAPATH from Early Adoption to Graduated [5/]



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Note: many commits done by Savoir-faire Linux in **2026** are **sponsored**

- Suse SLES support
- Support of Qualcomm IQ9

SEAPATH from Early Adoption to Graduated [6/]

- Have a public list of project adopters for at least the primary repo (e.g., ADOPTERS.md or logos on the project website) 
 - <https://github.com/seapath/.github/blob/main/ADOPTERS.md>

Areas the project could use help on

- Help to organise dedicated events (Workshop)
- Share REX on tools
 - Guidance on documentation (ex: move from confluence)
 - Tools to manage Roadmap
 - Tools to organize workshop (ex: brainstorming)
- Help on project management
 - Management of TSC & Voting process
 - Management of meetings

Feedback on working with LF Energy

- Good promotion of the project (social network & events)
- Lot of value to the project regarding life cycle maturity, openSSF badging and security audit
- Issues with SIG
 - Digital Substation SIG impact was very limited
 - Lot of issues to get SEAPATH project reviewed (last review done on Jan. 2025)
- Technical support is sometimes slow
 - ex: 6 months to have access to cloudflare for <https://releases.seapath.org>